# Logic Gates / තර්කන ද්වාර

මූලික ද්වාර Basic logic gate

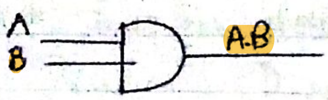
* AND
* OR
* NOT

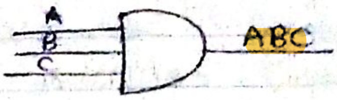
සංයෝජිත ද්වාර Combinational

* NAND
* NOR
* XOR
* XNOR

## Basic Logic Gates

### AND gate - [ගුණිතය]



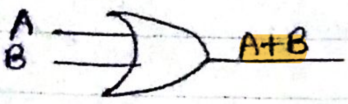




Truth table:-
සත්‍යතා වගුව

| A | B | AB |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

* AND operator produces a logical True only when both it's inputs are logical true.

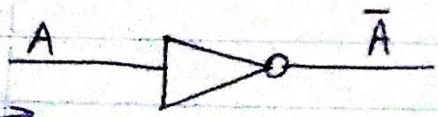
### OR Gate - [ඓක්‍යය]




| A | B | A+B |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

* OR operature produces a logical true output when any of input is logical True.

### NOT Gate [පූරකය]

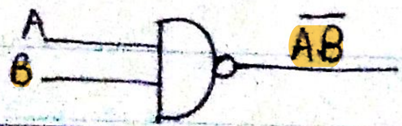



| A | $\bar{A}$ |
|---|---|
| 0 | 1 |
| 1 | 0 |

## Combination Gates

### NAND Gate

NOT + AND ⟶ NAND




| A | B | AB | $\overline{AB}$ |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 |

* NOT Operator ⇒ Perform a simple inverse

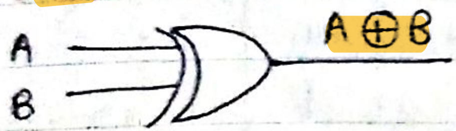
## NOR Gate

NOT + OR $\longrightarrow$ NOR


| A | B | A+B | $\overline{A+B}$ |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 |

* NOR is equivalent to an OR operator followed by a NOT operator

## XOR Gate




$\Rightarrow$ { $\oplus \Rightarrow A\bar{B} + \bar{A}B$ }

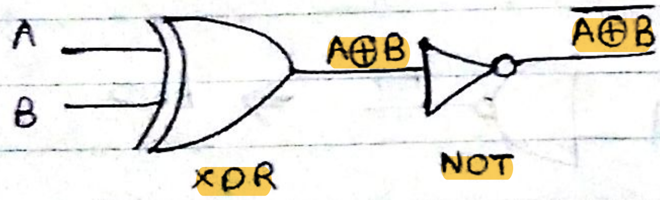
* XOR produces a logical true when either of its inputs is logical true.

| A | B | A + B | A ⊕ B |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 |

* සමාන ආදාන සඳහා output "0" වේ.
* අසමාන ආදාන සඳහා output "1" වේ.

## XNOR Gate

XOR + NOT $\longrightarrow$ XNOR




| A | B | A ⊕ B | $\overline{A \oplus B}$ |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

# Universal grade කියන්නේ මොනවද

$\left.\begin{array}{l} \text{NAND} \\ \text{NOR} \end{array}\right\}$ universal gate.

* මෙවා ඇලින් (universal gate) කරල දිගාර නිර්මාණය කළ හැ, ය.

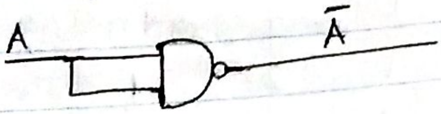
NAND ⟶ NOT

$\overline{AB}$

NAND →

| A | B | A.B | $\overline{AB}$ |
|---|---|---|---|
| 0 | 0 | 0 | 1 → (*) |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 → (*) |

* ආදානය මුලට එකම අගයන් ඇද යෙදූ විට. ප්‍රතිදානය NOT ආකාරයට ගත හැක.

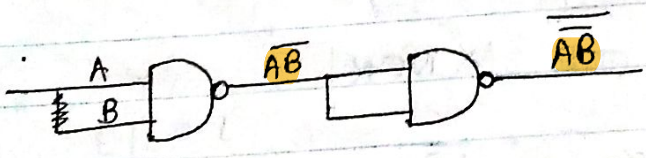


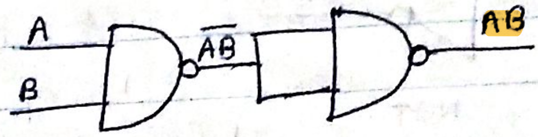

⇒ NAND → NOT

NAND ⟶ AND

$\overline{AB}$ ⟶ $AB$

$\overline{\overline{AB}}$ ⇒ $AB$




⇒ AND

NAND ⟶ OR

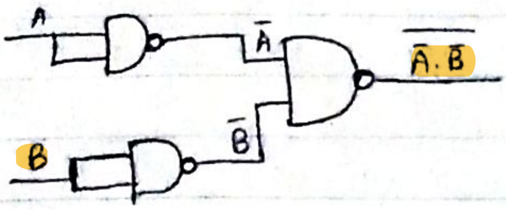
$\overline{AB}$ ⟶ $A+B$

$$\left.\begin{array}{l} \overline{AB} = \bar{A} + \bar{B} \\ \overline{\bar{A} + \bar{B}} = \bar{\bar{A}} \cdot \bar{\bar{B}} \end{array}\right\} \longrightarrow$$

$\overline{A \cdot B} \Rightarrow \bar{A} + \bar{B} \longrightarrow A+B$

$\Rightarrow \overline{\bar{A} + \bar{B}} = A+B$ //

$A+B = \overline{\bar{A} + \bar{B}} = \overline{\bar{A} \cdot \bar{B}}$



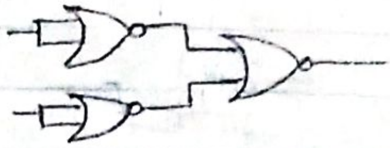

NOR $\longrightarrow$ AND

$\overline{A+B} \longrightarrow AB$

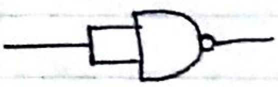
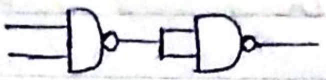
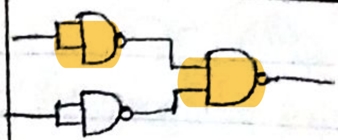
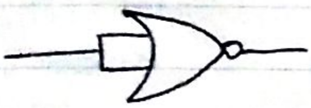
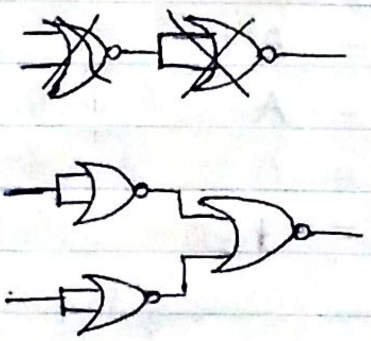
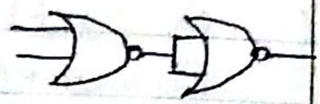
$\overline{A+B} = \bar{A} \cdot \bar{B}$ ~~← A~~

$\overline{\bar{A}\bar{B}} = AB$

$\overline{\bar{A} + \bar{B}} = AB$



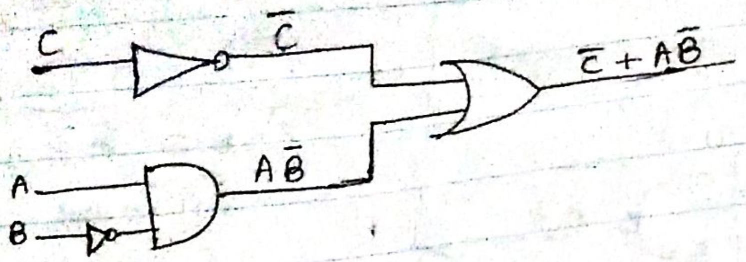
## Universal Gate Short notes

| | NOT | AND | OR |
|---|---|---|---|
| NAND |  |  |  |
| NOR |  |  |  |

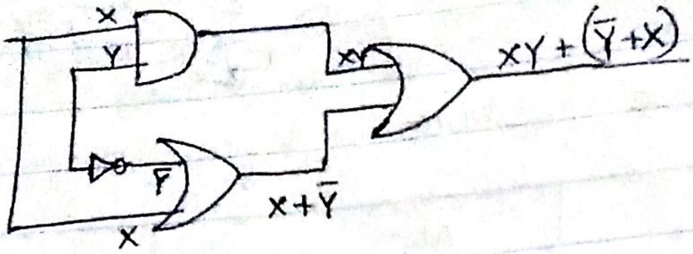
## Truth Table Short notes.

| A | B | AND | OR | NAND | NOR | ~~NXOR~~ XNOR | ~~XNOR~~ XOR |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 |

① $F = \overline{C} + A\overline{B}$




②




## Booleans laws

### Law of sum

① $A + 1 = 1$

② $A + A = A$

③ $A + 0 = A$

④ $A + \overline{A} = 1$

* $A \oplus B = A\overline{B} + \overline{A}B$

### Law of product

⑤ $A \times 1 = A$

⑥ $A \times A = A$

⑦ $A \times 0 = 0$

⑧ $A \times \overline{A} = 0$

### De morgan's Law

⑨ $\overline{A + B} = \overline{A} \cdot \overline{B}$

⑩ $\overline{A \cdot B} = \overline{A} + \overline{B}$

### Redundancy law

⑪ $X + \overline{X}Y = X + Y$

$X + \overline{X}Y = (X + \overline{X}) \cdot (X + Y)$

$A \times A = A$

$A \times A \times A = A$

$A + A + A = A$

### Distribution law

* $A + BC = (A+B) \cdot (A+$

① $AB + A\bar{B}$

$A(B+\bar{B})$

$A \times 1$

$= A$ //

② $(A+B)(A+\bar{B})$

$A + B\cdot\bar{B}$.

$A + 0$

$A$ //

$A\cdot A + A\bar{B} + BA + B\cdot\bar{B}$

$A + A\bar{B} + AB \quad 0$

$A + A(\bar{B}+B)$

$A + A\times 1 = A$ //

③ $\overline{A\cdot\overline{BC}}\,(A+BC)$

$(\bar{A} + \overline{\overline{BC}})(A+BC)$

$(\bar{A} + BC)(A+BC)$

$BC + A\cdot\bar{A}$

$BC + 0$

$BC$ //

④ $\bar{A}(A+B) + (B + \overline{\bar{A}\cdot A})(\bar{A}+\bar{B})$

$\bar{A}\cdot A + B\bar{A} + (B+\bar{A})(\bar{A}+\bar{B})$

$0 + B\bar{A} + (B+A)(\bar{A}+\bar{B})$

$B\bar{A} + B\bar{A} + B\bar{B} + A\bar{A} + A\bar{B}$

$B\bar{A} + B\bar{A} + 0 + 0 + A\bar{B}$

⑤ $\overline{(A+\bar{B})(\overline{A+B})}$

$\overline{(A+\bar{B})} + \overline{(\overline{A+B})}$

$\bar{A}\cdot\bar{\bar{B}} + \overline{\bar{A}\cdot\bar{B}}$

$\bar{A}\cdot B + \cancel{\bar{\bar{A}}\bar{\bar{B}}}\ \bar{\bar{A}} + \bar{\bar{B}}$

$\bar{A}\cdot B + A + B$

$B(\bar{A}+1) + A$

$B + A$ //

⑥ $ABC + A\bar{B}C + AB\bar{C}$

$= AC(B+\bar{B}) + AB\bar{C}$

$= AC(1) + AB\bar{C}$

$= A(C + B\bar{C})$

$= A(C+B)$

$= AC + AB$ //

④ $\overline{(A+B)(A+\bar{B})}$

$\overline{(A+B)} + \overline{(A+\bar{B})}$

$\bar{A}\cdot\bar{B} + \bar{A}\cdot\bar{\bar{B}}$

$\bar{A}\cdot\bar{B} + \bar{A}\cdot B$

$\bar{A}(\bar{B}+B)$

$\bar{A}\times 1$

$\bar{A}$ //

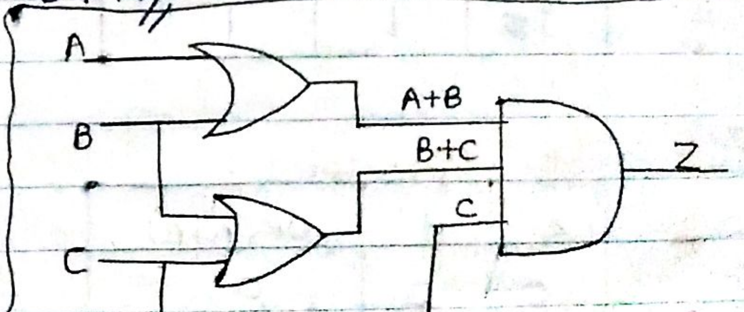
⑦ $ABC + A\bar{B}C + AB\bar{C} + \bar{A}BC$

$AB(C+\bar{C}) + AC(\bar{B}+B)$

$AB(1) + AC(1)$

$AB + AC$

$A(B+C) = AB + AC$ //



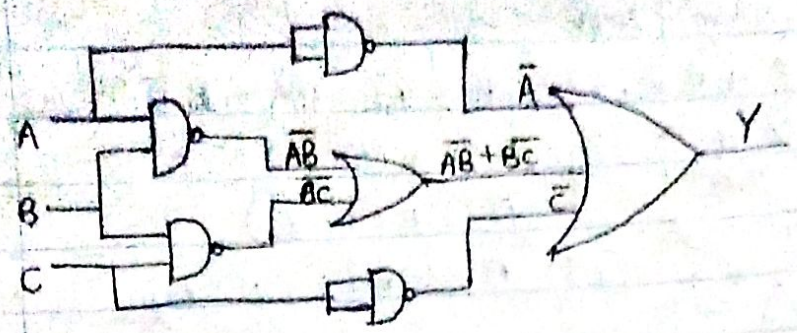

ප්‍රදාන ගණන = 3

ප්‍රතිදාන ගණන = $2^3 = 8$

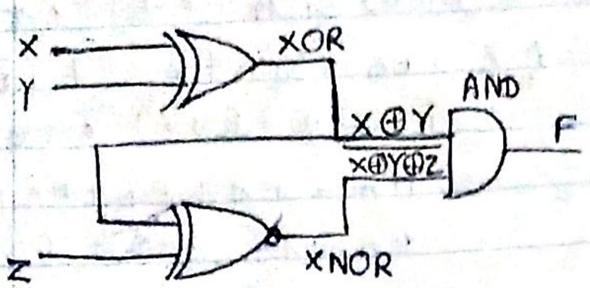
$Z = (A+B)\cdot(B+C)\cdot C$

Truth table ⇒

| A | B | C | A+B | B+C | Z |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 |




$\Rightarrow Y = \bar{A} + \overline{AB} + \overline{BC} + ...$




$\Rightarrow F = \overline{(X\oplus Y)\cdot(X\oplus Y\oplus Z)}$

$F = (X\oplus Y)\cdot(\overline{X\oplus Y\oplus Z})$

$F = (X\oplus Y)\cdot((\overline{X\oplus Y})\cdot \bar{Z})$

## Logical Operators.

XOR → (output)

### AND Operator.

| A | B | AND ~~(A+B)~~ AB |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

### NAND operator

| A | B | NAND ($\overline{AB}$) |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

### OR operator

| A | B | OR (A+B) |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

### NOR operator

| A | B | NOR ($\overline{A+B}$) |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

### NOT operator.

| A | ($\bar{A}$) NOT |
|---|---|
| 0 | 1 |
| 1 | 0 |

### XOR (Exclusive-OR) operator

| A | B | XOR ($A\oplus B$) |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

# Logical Function.

① $AB + A\bar{B}C + A\bar{B}\bar{C}$

$= AB + A\bar{B}\underbrace{(C + \bar{C})}$

$= A(AB) + A\bar{B}.1$

$= AB + A\bar{B}$

$= A\underbrace{(B + \bar{B})}_{1}$

$= A$ //

② $A + \bar{A}B$

$= \underbrace{(A+\bar{A})}.(A+B)$

$= 1.(A+B)$

$= A+B$ //

③ $A + \bar{A}\bar{B}$

$\underbrace{(A+\bar{A})}(A+\bar{B})$

$= 1.(A+\bar{B})$

$= (A+\bar{B})$ //

④ $\bar{A} + AB$

$= \underbrace{(\bar{A}+A)}\cdot(\bar{A}+B)$

$= 1(\bar{A}+B)$

$= (\bar{A}+B)$ //

⑤ $A\bar{B} + AB + \bar{A}B$

$= A\underbrace{(\bar{B}+B)} + \bar{A}B$

$= A\times 1 + \bar{A}B$

$= A + \bar{A}B$

$= \underbrace{(A+\bar{A})}_{1}.(A+B)$

$= A+B$ //

⑥ $A\bar{B} + AB + \bar{A}B + AB$

$= A\underbrace{(\bar{B}+B)} + B\underbrace{(\bar{A}+A)}$

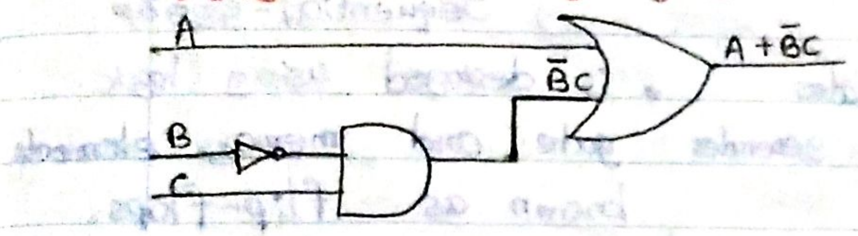
$= A\times 1 + B\times 1$

$= A + B$ //

⑦ Design $f(A, B, C)$

$= A + \bar{B}C$

Using Basic Logic gate.

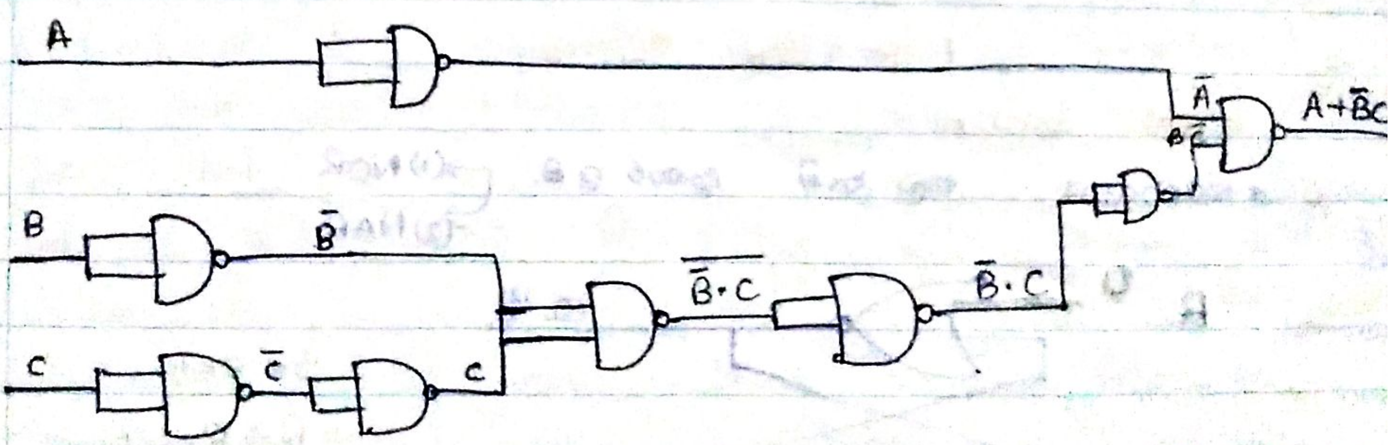


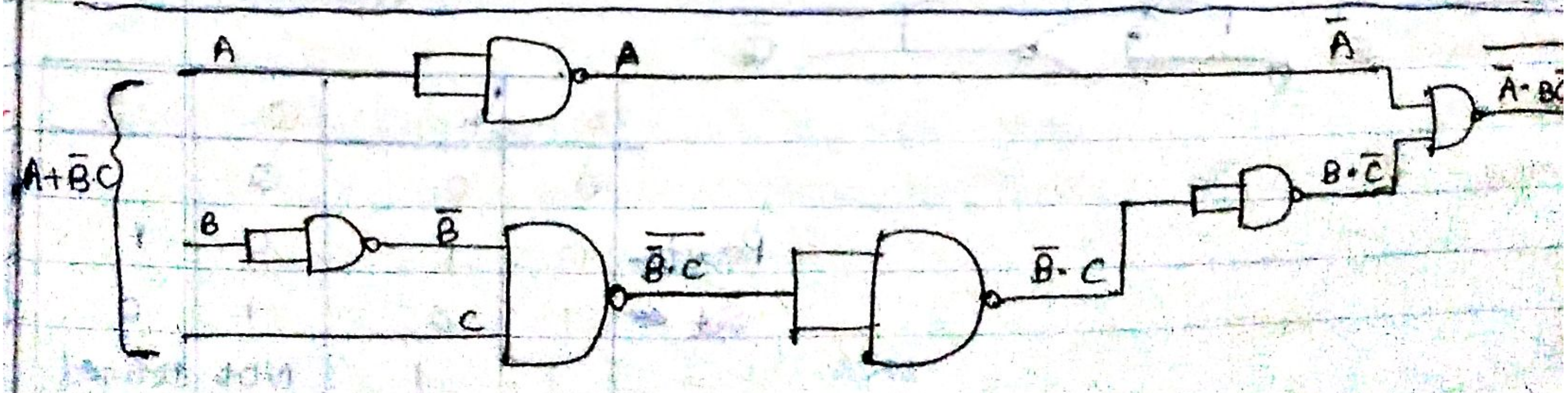

$\overline{\bar{A}\cdot\overline{\bar{B}C}}$

$\bar{\bar{A}} + \overline{\overline{\bar{B}C}}$

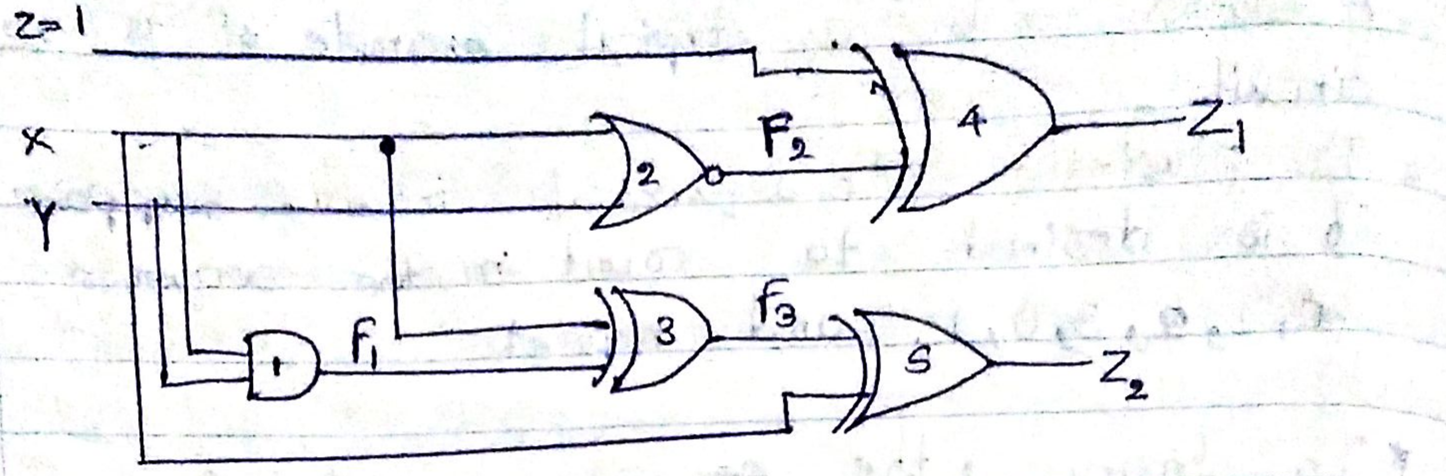
$= A + \bar{B}C$ //

Using universal Logic gate.



# Combinational logic circuit




input $\Rightarrow$ $X, Y, Z$

Output $\rightarrow$ $Z_1, Z_2$

$F_1 = X \cdot Y$

$F_2 = \overline{X+Y}$

$Z_1 = 1 \oplus F_2$

$Z_1 = 1 \oplus \overline{X+Y}$

$Z_1 = \overline{F_2}$

~~$Z_1 = X+Y$~~

$Z_1 = X+Y$

$F_3 = X \oplus XY$

$Z_2 = F_3 \oplus X$

$Z_2 = (X \oplus XY) \oplus X$

$Z_2 = X \oplus (X(1 \oplus Y))$

$= X \oplus X\overline{Y}$

$=$ Since $1 \oplus Y = \overline{Y}$

$= X(1 \oplus \overline{Y})$

$Z_2 = XY$ // Since $1+\overline{Y} = Y$ //

$Z_2 = X \oplus (X \oplus XY) = X \oplus X \oplus XY$

$= 0 \oplus XY = XY$ //

## Design of a combination Circuit

Truth table:

| Input | | Output | |
|---|---|---|---|
| X | Y | $Z_1$ | $Z_2$ |
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 |

### Operator

① **Arithmatics**
- Addition
- Substraction
- Multiplication
- Division

② **Comparson**
- $<$ less than
- $>$ Greater than
- $\leq$ less than or equal
- $\geq$ Greater than or equal
- $=$ equal
- $\neq$ not equal

③ **Logic**
- ~~AND~~ AND
- OR
- NOT
- NOR
- NAND
- X-OR
- ~~XOI~~ X-NOR

**1- bit addition**

$0 + 0 = 0$
$0 + 1 = 1$
$1 + 0 = 1$
$1 + 1 = 0$

**1-bit Substraction**

$0 - 0 = 0$
$0 - 1 = 1$
$1 - 0 = 1$
$1 - 1 = 0$

⊛ XOR ⟵ both 1-bit addition and 1-bit Substraction

### Logic function

① What is function?
Relationship between two variables.

a, b, c ⇒ logic variable

$f(a,b,c) = abc + \bar{a}b\bar{c}$ ⟶ logic function

Logic function $\xrightarrow{\text{Domain}}$ 0, 1

Simplify following boolean expression.

① $A(\bar{A}+B)\cdot C = \underbrace{A\cdot\bar{A}C}_{0} + A\cdot BC$

$+ ABC = ABC$

② $\overline{AB}\cdot(A+B) =$

$\cancel{\bar{A}BA + \bar{A}\underbrace{B\bar{B}}_{0}}$

$(\bar{A}+\bar{B})\cdot(A+B)$

$= \underbrace{A\bar{A}}_{0} + \bar{A}B + \bar{B}A + \underbrace{B\bar{B}}_{0}$

$= \bar{A}B + \bar{B}A = (A\oplus B)$

③ $\overline{(A+B)} + \bar{B}C = \bar{A}\bar{B} + \bar{B}C$

$= \bar{B}(\bar{A}+C)$

④ $x\cdot y\,(x+y) + x = xy\cdot x + x\cdot y\cdot y + x$

$= xy + xy + x$

$= xy + x$

$= x\underbrace{(y+1)}_{1} = x$

⑤ $\bar{P} + \bar{P}(Q+x) + xQ$

$= \bar{P} + \bar{P}Q + \bar{P}x + xQ$

$= \bar{P}(1+Q) + x(\bar{P}+Q)$

$= \bar{P} + \bar{P}x + xQ$

$= \bar{P}\underbrace{(1+x)}_{1} + xQ$

$= \bar{P} + xQ$

⑥

⑥ $\bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D}$

$= \bar{A}\bar{B}(\bar{C}D + C\bar{D}) + A\bar{B}\bar{D}\underbrace{(\bar{C}+C)}_{1}$

$= \bar{A}\bar{B} + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{D}$

$= \bar{B}(\bar{A}\bar{C}D + \bar{A}C\bar{D} + A\bar{D})$

$= \bar{B}(\bar{A}\bar{C}D + \bar{D}(\bar{A}C + A))$

$= \bar{B}(\bar{A}\bar{C}D + \bar{D}\underbrace{(A+\bar{A})}_{1}\cdot(A+C))$

$= \bar{B}(\bar{A}\bar{C}D + \bar{D}A + \bar{D}C)$

# Karnaugh Map(K-Map)

Input ස 2 → A, B

| A | B | F |
|---|---|---|
| 0 | 1 | 0 |
| 1 | 0 | 1 * |
| 0 | 0 | 1 * |
| 1 | 1 | 0 |

4 ක් නිසා
හා K-Map එකෙ කොටු 4

K-Map

Origin එක

| | |
|---|---|
| | |
| | |

* Orgin එකේ ඇතුල කොටුව → ඇතිවා '0' වේ.

| B \ A | | 0 | 1 |
|---|---|---|---|
| $\bar{B}$ | 0 | 1 | 1 |
| B | 1 | | |

* කෙසේ නමුත් මූ k-map කෑදීමට ඇතිවීම,

(1) Boolean ප්‍රකාශනය ලියා ගත යුතුය.

- Output එක = 1 වන අවස්ථාව බලා Boolean ප්‍රකාශනය සාදයි

$$F = A.\bar{B} + \bar{A}B.$$

මෙය k-map එකේ ලියාගනී.

## K-Map Group කිරීමේ නීති

① අනුලෝම 1 ඇති කොටු 2කේ බල වල

① $F = \bar{A}B + AB$

| B \ A | 0 | 1 |
|---|---|---|
| $\bar{B}$ 0 | 0 | 0 |
| B 1 | 1 | 1 |

$F = B(\bar{A} + A)$

$= B \cdot 1$

$F = B$ //

② $F = \bar{A}\bar{B} + \bar{A}B + AB$

| B \ A | 0 | 1 |
|---|---|---|
| $\bar{B}$ 0 | 1 | 0 |
| B 1 | 1 | 1 |
| | $\bar{A}$ | A |

~~$\bar{A}\bar{B} + \bar{A}B +$~~

$\bar{A}\bar{B} + \bar{A}B$

$\bar{A}B + AB$

$F = \bar{A} + B$ //

③ $F = A\bar{B} + AB + \bar{A}\bar{B}$

| B \ A | 0 | 1 |
|---|---|---|
| $\bar{B}$ 0 | 1 | 1 |
| B 1 | 0 | 1 |
| | $\bar{A}$ | A |

$= \bar{A}\bar{B} + AB$

$\quad A\bar{B} \quad A\bar{B}$

$F = \bar{B} + A$ //

④ $F = \bar{A}BC + ABC + \bar{A}\bar{B}C + \bar{A}B\bar{C}$

| C \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 |
| | $\bar{A}\bar{B}$ | $\bar{A}B$ | $AB$ | $A\bar{B}$ |

$= \bar{A}\bar{B}C + \bar{A}B\bar{C} + ABC$

$\quad \bar{A}BC \quad \bar{A}BC \quad \bar{A}BC$

$= \bar{A}C + \bar{A}B + BC$ //

⑤

| A | B | C | $F_1$ |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

| C \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| $\bar{C}$ 0 | 1 | 1 | 1 | 0 |
| C 1 | 0 | 1 | 0 | 1 |
| | $\bar{A}\bar{B}$ | $\bar{A}B$ | $AB$ | $A\bar{B}$ |

$F_1 = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + AB\bar{C} + A\bar{B}C$

$\quad \bar{A}B\bar{C} \quad \bar{A}BC \quad \bar{A}B\bar{C}$

$F_1 = \bar{A}\bar{C} + \bar{A}B + B\bar{C} + A\bar{B}C$

⑥

| A | B | C | $F_2$ |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |

| C \ AB | | $\bar A\bar B$ 00 | $\bar A B$ 01 | $AB$ 11 | $A\bar B$ 10 |
|---|---|---|---|---|---|
| $\bar C$ | 0 | 1 | 1 | 1 | 0 |
| $C$ | 1 | 1 | 0 | 1 | 1 |

$\bar A\bar B\bar C + \bar A B\bar C + AB\bar C + ABC$

$\bar A\bar B C \quad \bar A\bar B\bar C \quad ABC \quad A\bar B C$

$\bar A\bar B + \bar A\bar C + AB + AC$

| C \ AB | | 00 | 01 | 11 | 10 |
|---|---|---|---|---|---|
| $\bar C$ | 0 | 1 | 1 | 1 | 0 |
| $C$ | 1 | 1 | 0 | 1 | 1 |
| | | $\bar A\bar B$ | $\bar A B$ | $AB$ | $A\bar B$ |

⑦ $F = \bar A BC + AC\bar D + \bar C\bar D$

| CD \ AB | | 00 | 01 | 11 | 10 |
|---|---|---|---|---|---|
| $\bar C\bar D$ | 00 | 1 | 1 | 1 | 1 |
| $\bar C D$ | 01 | 0 | 0 | 0 | 0 |
| $CD$ | 11 | 0 | 1 | 0 | 0 |
| $C\bar D$ | 10 | 0 | 1 | 1 | 1 |
| | | $\bar A\bar B$ | $\bar A B$ | $AB$ | $A\bar B$ |

$F = \bar A BC + AC\bar D + \bar C\bar D$

$\quad\;\; 011 \quad\; 110 \quad\; 00$

$\bar A\bar B\bar C\bar D + \bar A BC\bar D + AB C\bar D$

$\bar A B\bar C\bar D \quad \bar A BCD \quad A\bar B C\bar D$

$AB\bar C\bar D$

$A\bar B\bar C\bar D$

$F = \bar C\bar D + \bar A BC + AC\bar D$ //

⑧ $F = \bar A + B\bar C\bar D + A\bar C\bar D + \bar B\bar C D + \bar B$

| CD \ AB | | $\bar A\bar B$ 00 | $\bar A B$ 01 | $AB$ 11 | $A\bar B$ 10 |
|---|---|---|---|---|---|
| $\bar C\bar D$ | 00 | 1 | 1 | 1 | 1 |
| $\bar C D$ | 01 | 1 | 1 | 0 | 1 |
| $CD$ | 11 | 1 | 1 | 0 | 1 |
| $C\bar D$ | 10 | 1 | 1 | 0 | 1 |

$\bar A\bar B\bar C\bar D + \bar A B\bar C\bar D + A\bar B\bar C\bar D$

$\bar A\bar B\bar C D \quad AB\bar C\bar D \quad A\bar B\bar C D$

$\bar A\bar B CD \quad AB\bar C\bar D \quad A\bar B CD$

$\bar A\bar B C\bar D \quad A\bar B\bar C\bar D \quad A\bar B C\bar D$

$\bar A B\bar C\bar D$

$\bar A B\bar C D$

$\bar A BCD$

$\bar A BC\bar D$

$= \bar A + \bar B + A\bar B$ //

$\cancel{\bar A + \bar C\bar D + A\bar B}$

(9)

| A | B | C | D | F |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 |

| CD \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| $\bar C\bar D$ 00 | 0 | 0 | 0 | 0 |
| $\bar C D$ 01 | 0 | 1 | 1 | 0 |
| $CD$ 11 | 1 | 1 | 0 | 1 |
| $C\bar D$ 10 | 1 | 0 | 0 | 0 |
| | $\bar A\bar B$ | $\bar A B$ | $AB$ | $A\bar B$ |

$\bar A B C\bar D + \bar A B \bar C D + A B \bar C D + A\bar B C D$

$\bar A\bar B C D \quad \bar A B C D$

$\bar A\bar B C + \bar A B C + A\bar B C$

([illegible]) $\bar B C D + \bar A\bar B C + \bar A B D + B\bar C D$

Theory ( Exam )

# Half Adder

( Input 2 ක් එකතු / binary සංඛ්‍යා 2 ක එකතුව )

$$\begin{array}{r} 1 \\ +\ 1 \\ \hline 10 \end{array}$$

$1 + 1 = 2$ යි.

Binary වලදී, $2^3$ $2^2$ $2^1$ $2^0$

8 4 2 1

1 = 0 0 0 1

2 → 1 0

Half adder මගින්දී,

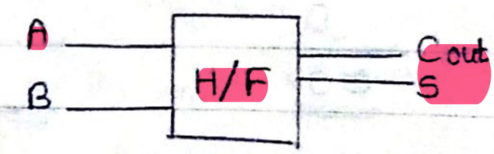
$$\begin{array}{r} 1 \\ +\ 1 \\ \hline 10 \end{array} \qquad \begin{array}{r} 0 \\ +\ 0 \\ \hline 00 \end{array} \qquad \begin{array}{r} 0 \\ +\ 1 \\ \hline 01 \end{array}$$

Carry → 1, Sum → 0

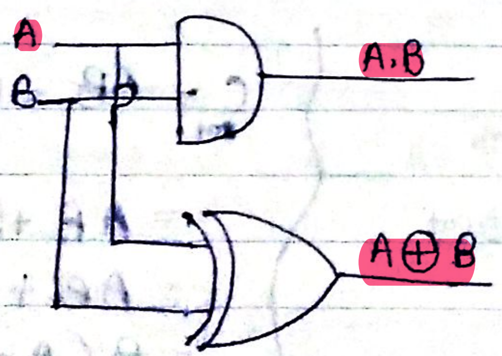
## Truth Table :-

| A | B | carry | Sum |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 |

carry ← AND, Sum ← XOR




## Logic gate for Half Adder.




Carry → AND } Half Adder.
Sum → XOR

# Full Adder

( [illegible] )

$A \oplus B \oplus C$

$(A \cdot B) + (A \cdot C) + (B \cdot C)$

Truth table

| A | B | C | Carry | Sum |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 |

Carry

$$\begin{array}{r} \boxed{1}\phantom{00} \\ 101 \\ +\,011 \\ 001 \\ \hline 100\boxed{1} \end{array}$$

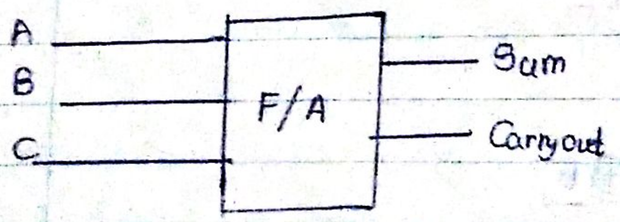
Sum

Carry

$$\begin{array}{r} 0 \\ +\,0 \\ 0 \\ \hline 00 \end{array} \quad \begin{array}{r} \boxed{0}\ \text{Carry} \\ 0 \\ +\,0 \\ 1 \\ \hline \boxed{1} \end{array} \quad \begin{array}{r} 1\ \text{Carry} \\ 0 \\ 1 \\ +\,1 \\ \hline \boxed{0} \end{array} \quad \begin{array}{r} 1\ \text{Carry} \\ 1 \\ 1 \\ +\,1 \\ \hline \boxed{1} \end{array}$$

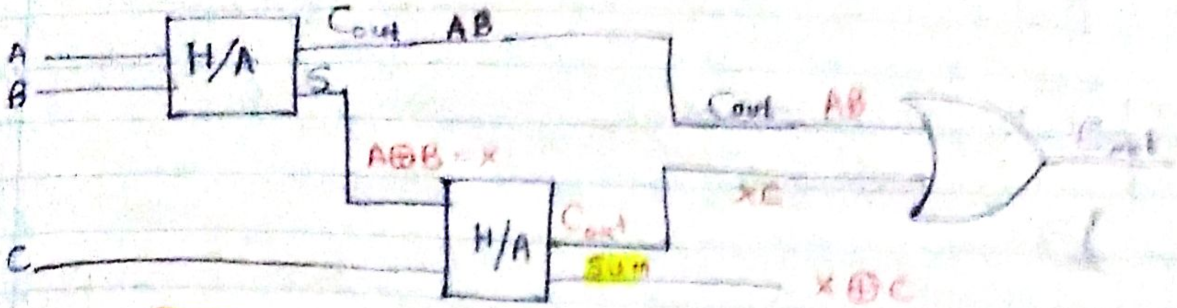
Sum  Sum  Sum  Sum

full adder { Carry $\rightarrow$ $A \cdot B + B \cdot C + A \cdot C$
Sum $\rightarrow$ $A \oplus B \oplus C$




$$\begin{aligned}
C_{out} &= AB + (A \oplus B)C \\
&= AB + (\bar{A}B + A\bar{B})C \\
&= AB + \bar{A}BC + A\bar{B}C \\
&= B(A + \bar{A}C) + A\bar{B}C \\
&= B(A + C) + A\bar{B}C \\
&= BA + BC + A\bar{B}C \\
&= A(B + \bar{B}C) + BC \\
&= A(B + C) + BC \\
&= AB + AC + BC
\end{aligned}$$

Full Adder using Half adder



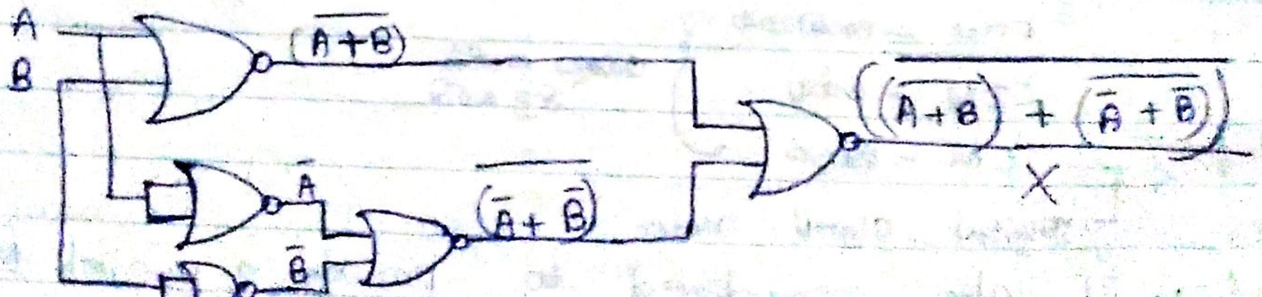

$$\text{Sum} = X \oplus C$$
$$= A \oplus B \oplus C$$

$$C_{out} = AB + XC$$
$$= AB + (A \oplus B)C \quad //$$

$$X = A \oplus B$$
$$= A\bar{B} + B\bar{A}$$
$$= A\bar{B} + A\bar{A} + B\bar{A} + B\bar{B}$$
$$= A(\bar{A}+\bar{B}) + B(\bar{A}+\bar{B})$$
$$X = (\bar{A}+\bar{B})(A+B)$$

$$\overline{\bar{X}} = \overline{\overline{(\bar{A}+\bar{B})(A+B)}}$$
$$X = \left(\overline{\overline{(\bar{A}+\bar{B})} + \overline{(A+B)}}\right)$$
$$\cancel{= \overline{(\bar{A}+\bar{B})} \cdot A+B}$$
$$X = \left(\overline{\overline{(\bar{A}+\bar{B})} + \overline{(A+B)}}\right)$$

$$\bar{C}_{out} = AB$$
$$\overline{C_{out}} = \overline{AB}$$
$$= \overline{\bar{A}+\bar{B}}$$



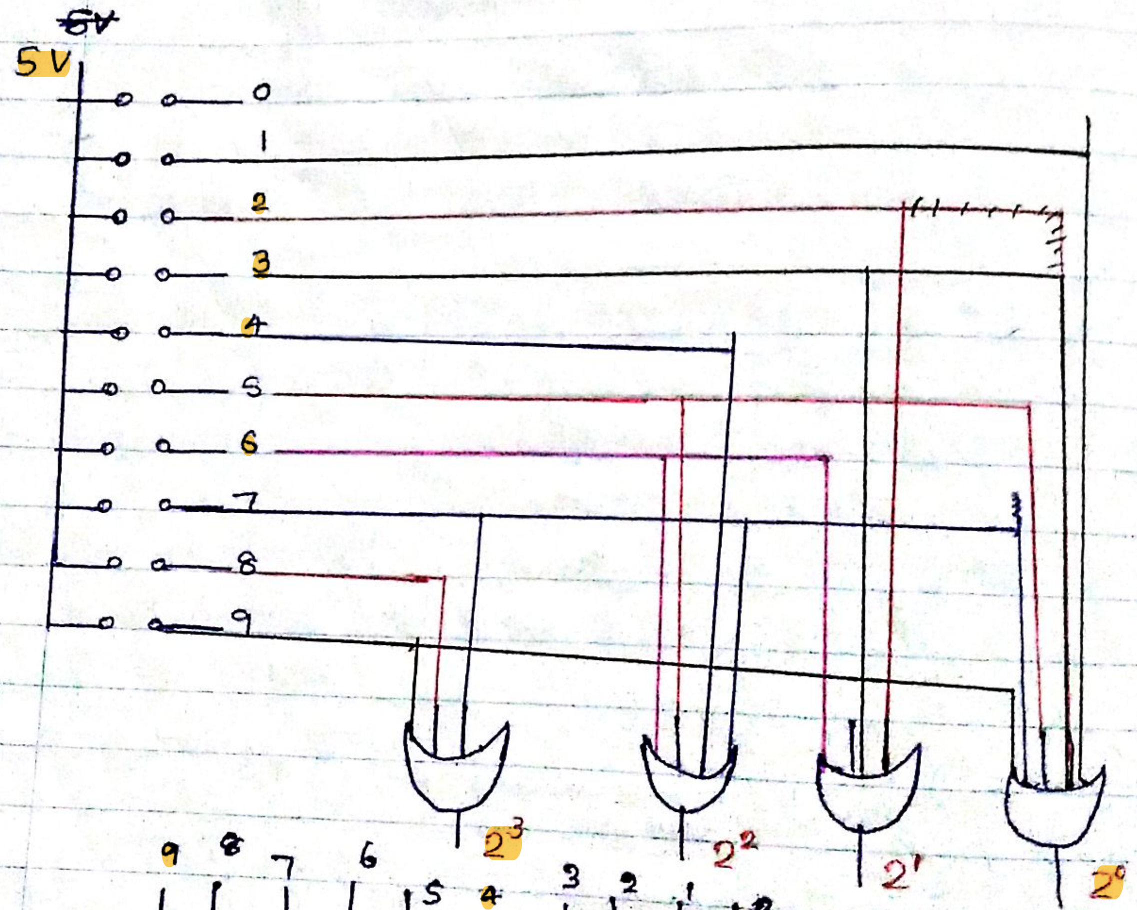
# Encoder (D2B)

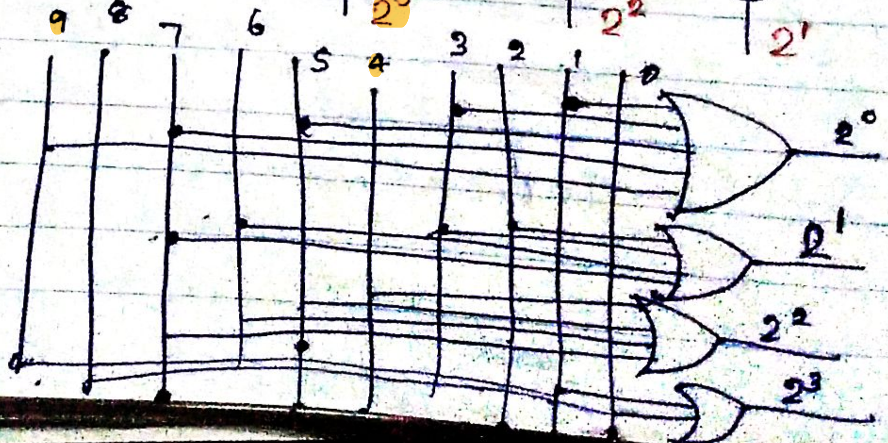
Decimal to Binary

(Decimal to BCD)

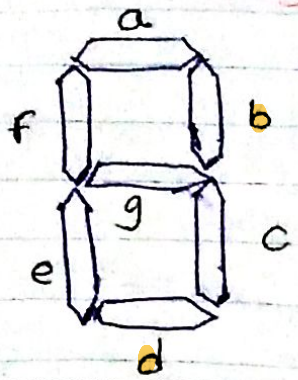
| Decimal digit | Binary $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 |
| 2 | 0 | 0 | 1 | 0 |
| 3 | 0 | 0 | 1 | 1 |
| 4 | 0 | 1 | 0 | 0 |
| 5 | 0 | 1 | 0 | 1 |
| 6 | 0 | 1 | 1 | 0 |
| 7 | 0 | 1 | 1 | 1 |
| 8 | 1 | 0 | 0 | 0 |
| 9 | 1 | 0 | 0 | 1 |

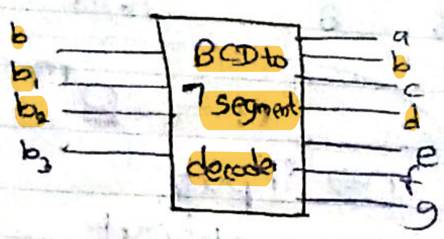
Decimal digits දශම සංඛ්‍යාවක් binary digits 4 කින් දක්වනු ලැබේ, 9 දක්වා digit 4 ක් අවශ්‍ය කරයි



# Seven Segment display Decoder




7 segment ⇒ create to display decimal number

| | $b_3$ | $b_2$ | $b_1$ | $b_0$ | a | b | c | d | e | f | g |
|---|---|---|---|---|---|---|---|---|---|---|---|
| (0) | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| (1) | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| (2) | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 |
| (3) | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| (4) | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 |
| (5) | 0 | 1 | 0 | 1 | 1 | 0 | 1 | [illegible] | 0 | 1 | 1 |
| (6) | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 |
| (7) | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| (8) | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| (9) | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | [illegible] |

# Multiplexer

* Multiplexer → Combinational circut.
  → is simply a DATA SELECTOR.


## Advantages

1. Reduces no of wires.
2. Reduces circuit complexity & cost.
3. Implementation of various circuits using M

## Octal to binary encoder

* Input 8 and Output 3.

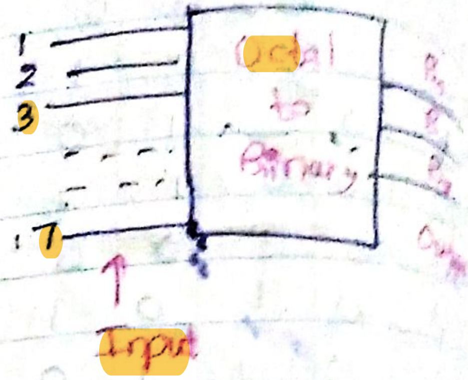
$$n = 2^m$$

$n \Rightarrow$ input

$m \Rightarrow$ output

$$8 = 2^3$$

| Input | $B_2$ | $B_1$ | $B_0$ |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 |
| 2 | 0 | 1 | 0 |
| 3 | 0 | 1 | 1 |
| 4 | 1 | 0 | 0 |
| 5 | 1 | 0 | 1 |
| 6 | 1 | 1 | 0 |
| 7 | 1 | 1 | 1 |

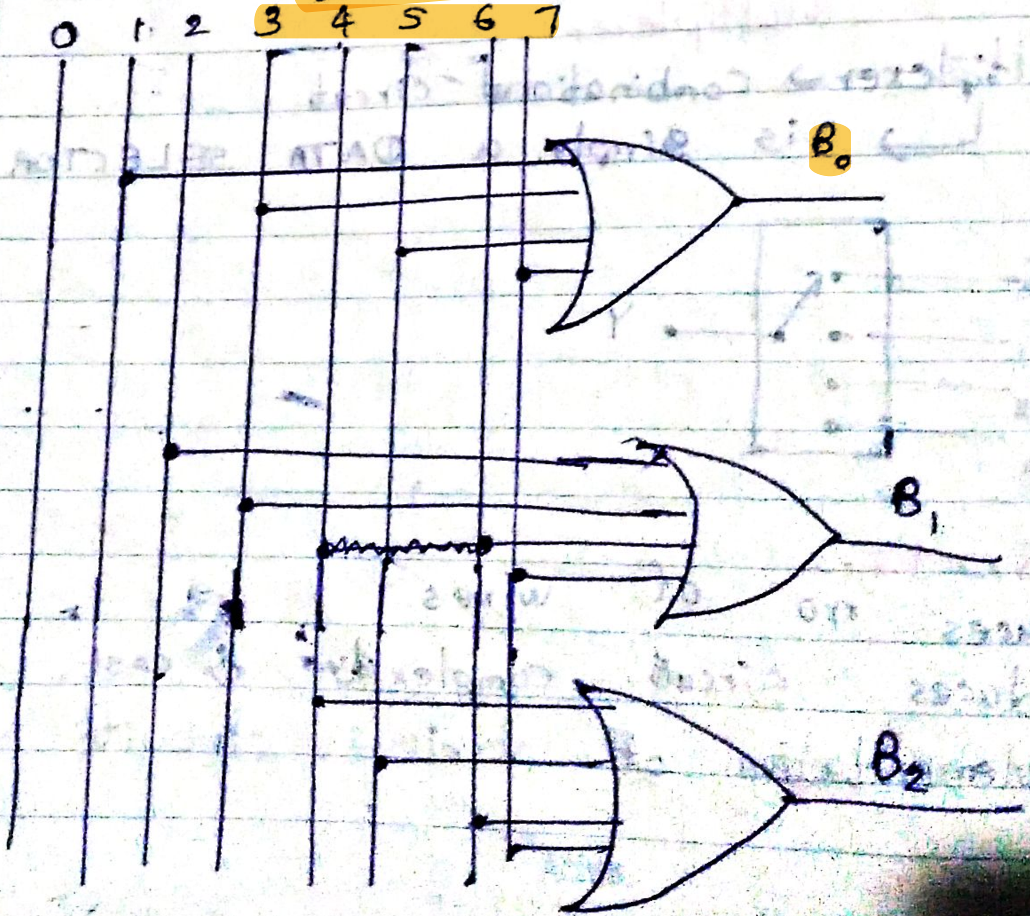



$B_0 \Rightarrow 1+3+5+7$

~~$B_1 \Rightarrow 2+3+4+5+6+7$~~

$B_1 \Rightarrow 2+3+6+7$

$B_2 \Rightarrow 4+5+6+7$

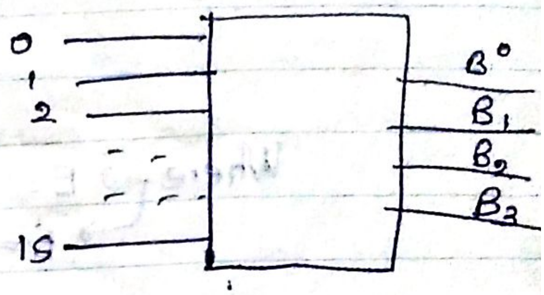


# Hexadecimal to binary ~~Decoder~~. Encoder

$\boxed{16} = 2^{4}$ ← output

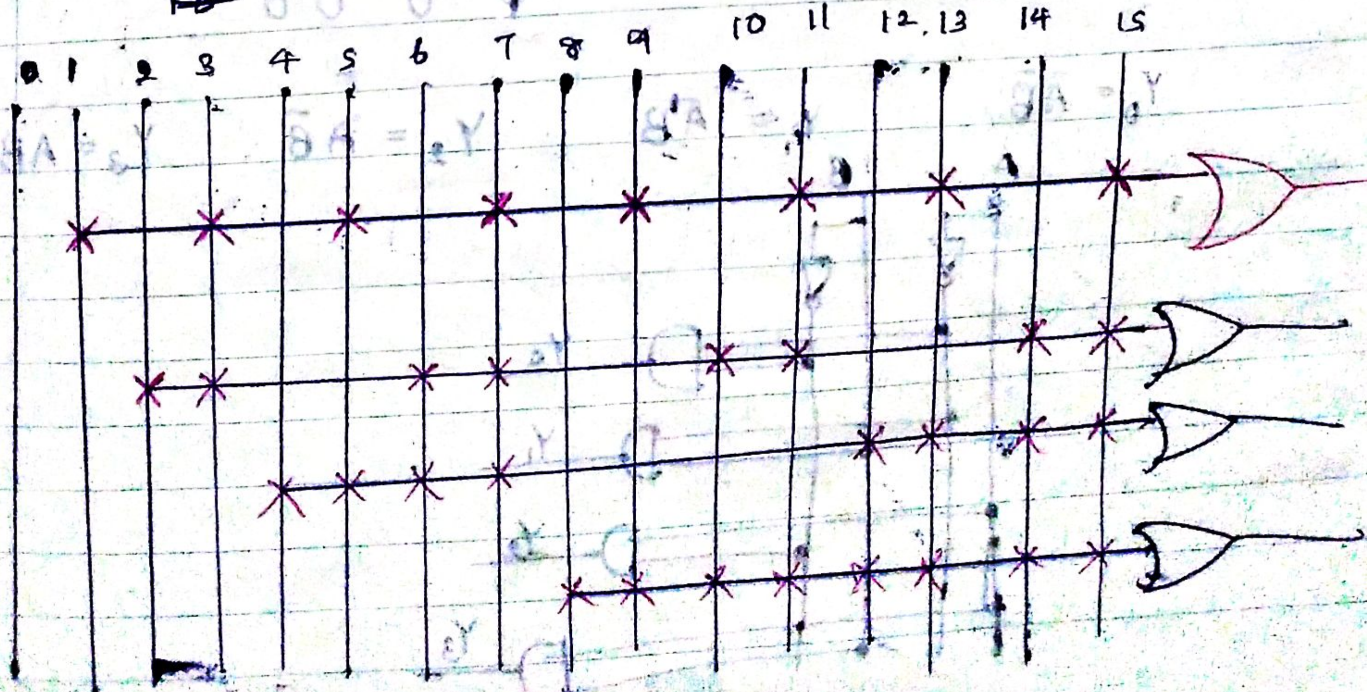
↑ Input

| Input | $B_3$ | $B_2$ | $B_1$ | $B_0$ |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 |
| 2 | 0 | 0 | 1 | 0 |
| 3 | 0 | 0 | 1 | 1 |
| 4 | 0 | 1 | 0 | 0 |
| 5 | 0 | 1 | 0 | 1 |
| 6 | 0 | 1 | 1 | 0 |
| 7 | 0 | 1 | 1 | 1 |
| 8 | 1 | 0 | 0 | 0 |
| 9 | 1 | 0 | 0 | 1 |
| A_10 | 1 | 0 | 1 | 0 |
| B_11 | 1 | 0 | 1 | 1 |
| C_12 | 1 | 1 | 0 | 0 |
| D_13 | 1 | 1 | 0 | 1 |
| E-14 | 1 | 1 | 1 | 0 |
| F-15 | 1 | 1 | 1 | 1 |
| ~~16~~ | | | | |




$B_0 = 1+3+5+7+9+11+13+15.$

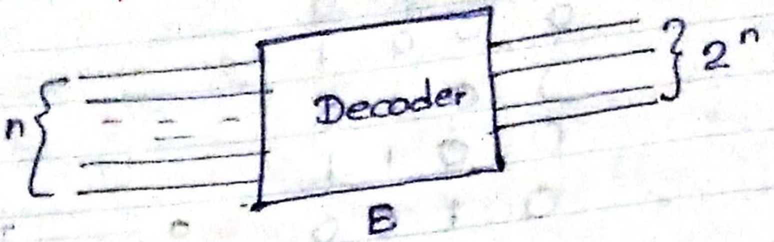
$B_1 = 2+3+6+7+10+11+14+15$



# Decoder

* Decoder is a multi input multi output logic circuit which decodes n inputs into $2^n$ possible outputs.

* If n input $\xrightarrow{\text{then}}$ Output $2^n$

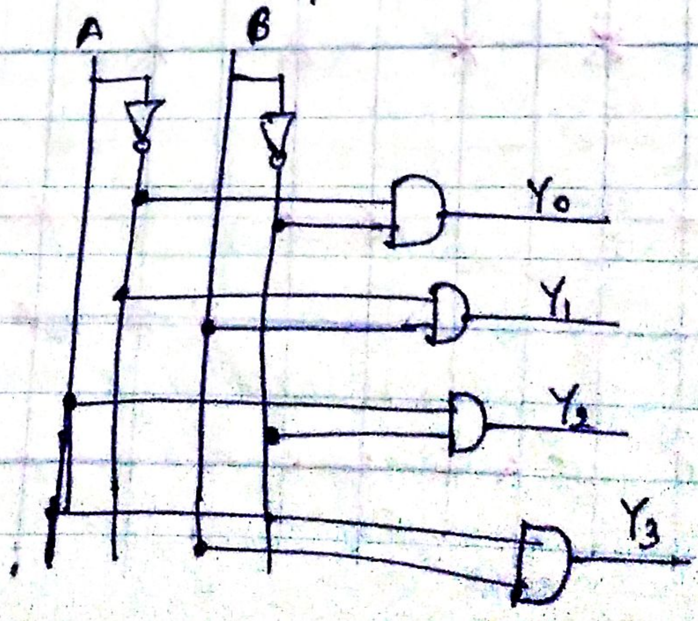



Where
- E = 0 → Decorder is disabled
- E = 1 → Decorder is enabled.

## 2 to 4 Decorder

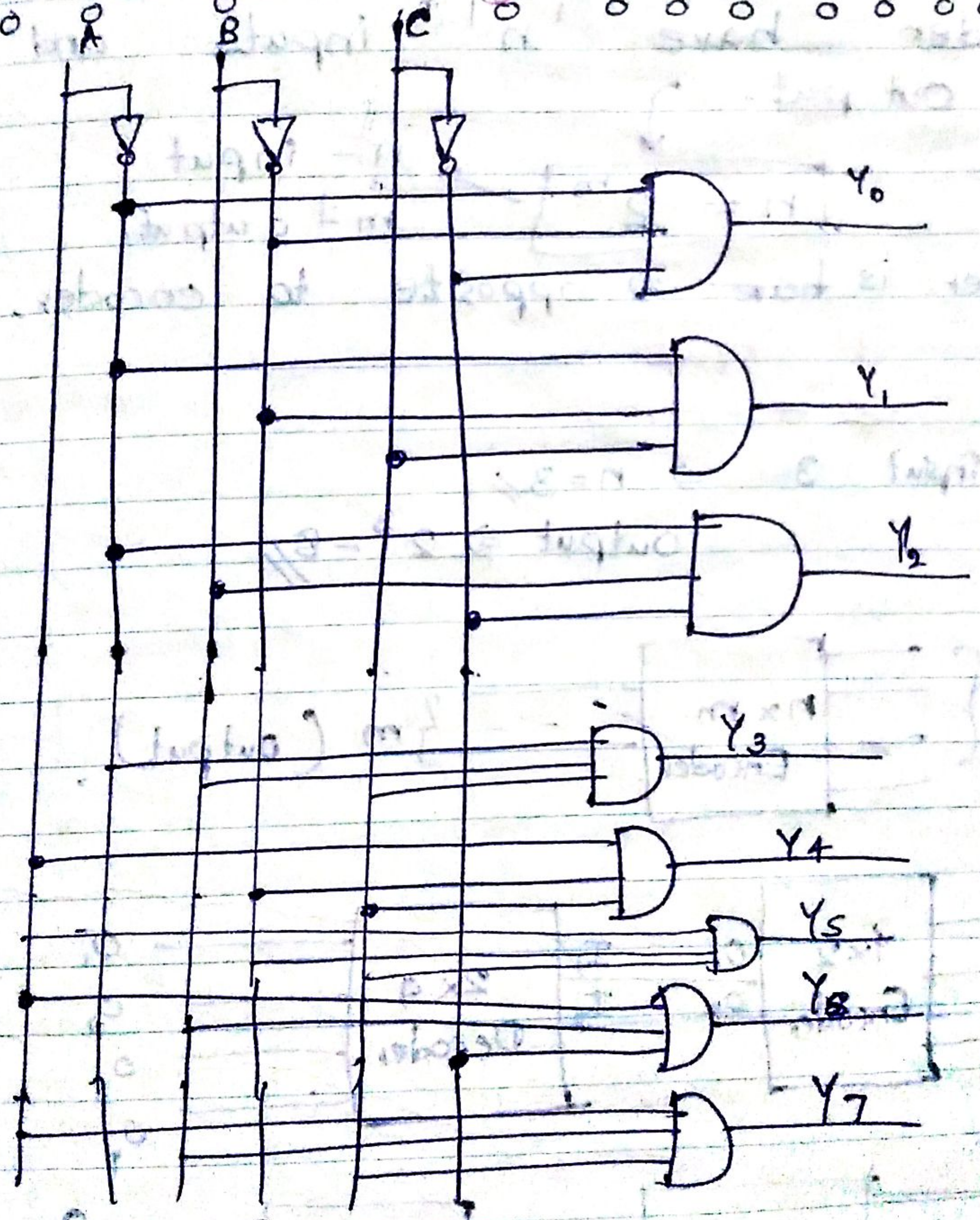
| E | A | B | $Y_3$ ($2^3$) | $Y_2$ ($2^2$) | $Y_1$ ($2^1$) | $Y_0$ ($2^0$) |
|---|---|---|---|---|---|---|
| 0 | X | X | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| | 0 | 1 | 0 | 0 | 1 | 0 |
| | 1 | 0 | 0 | 1 | 0 | 0 |
| | 1 | 1 | 1 | 0 | 0 | 0 |

$Y_0 = \bar{A}\bar{B}$ $Y_1 = \bar{A}B$ $Y_2 = A\bar{B}$ $Y_3 = AB$

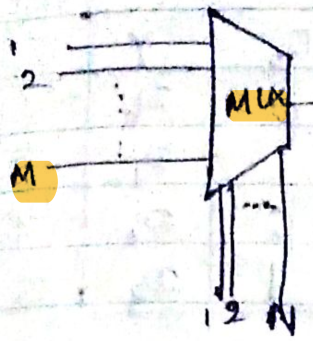


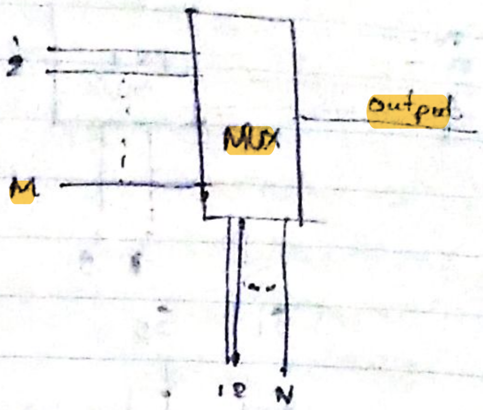
# 2 to 8 Decoder

| E | A | B | C | | $Y_7$ ($2^7$) | $Y_6$ ($2^6$) | $Y_5$ ($2^5$) | $Y_4$ ($2^4$) | $Y_3$ ($2^3$) | $Y_2$ ($2^2$) | $Y_1$ ($2^1$) | $Y_0$ ($2^0$) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0 | 0 | 0 | $\bar{A}\bar{B}\bar{C}$ | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | $\bar{A}\bar{B}C$ | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | $\bar{A}B\bar{C}$ | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | $\bar{A}BC$ | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | $A\bar{B}\bar{C}$ | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | $A\bar{B}C$ | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 0 | $AB\bar{C}$ | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | $ABC$ | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |



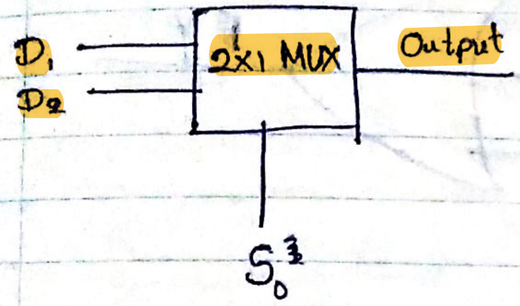
## Multiplexer → Boolean function Implementation



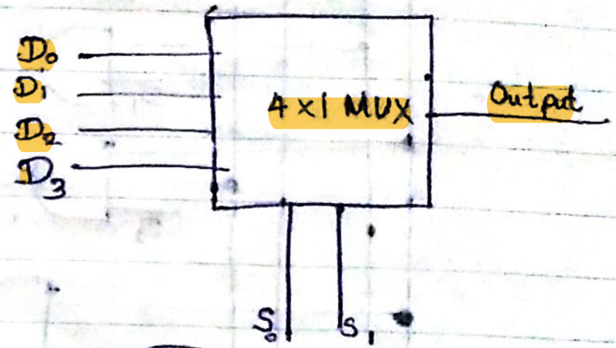

input → $M = 2^N$ ← selection

### 2x1 MUX



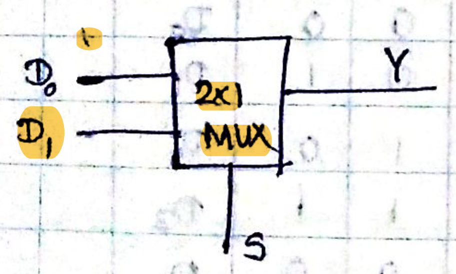

$\therefore 2 = 2^1$

### 4x1 MUX




$\therefore 4 = 2^2$

### 2X1 Multiplexer




Truth table ⇒

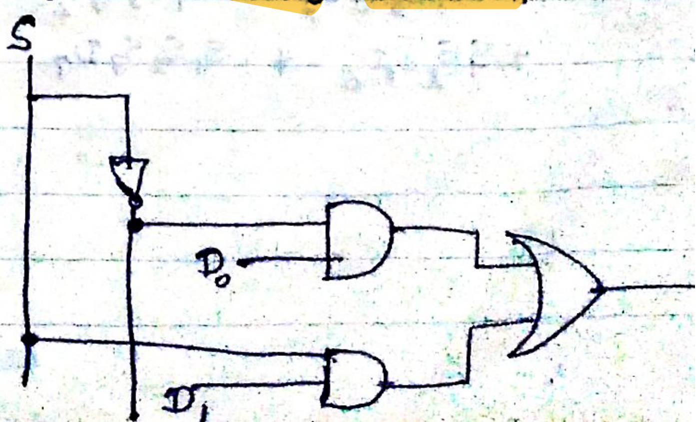
| S | Y |
|---|---|
| 0 | $D_0$ |
| 1 | $D_1$ |

circuit

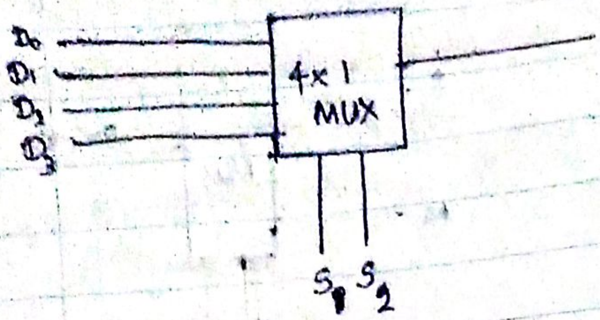
~~$Y = S_0 D$~~

$$Y = \bar{S} D_0 + S D_1$$

| $D_0$ | $D_1$ | S | Y |
|---|---|---|---|
| 0 | 0 | 0 | $D_0$ |
| 0 | 0 | 1 | $D_1$ |
| 0 | 1 | 0 | $D_0$ |
| 0 | 1 | 1 | $D_1$ |
| 1 | 0 | 0 | $D_0$ |
| 1 | 0 | 1 | $D_1$ |
| 1 | 1 | 0 | $D_0$ |
| | | | $D_1$ |

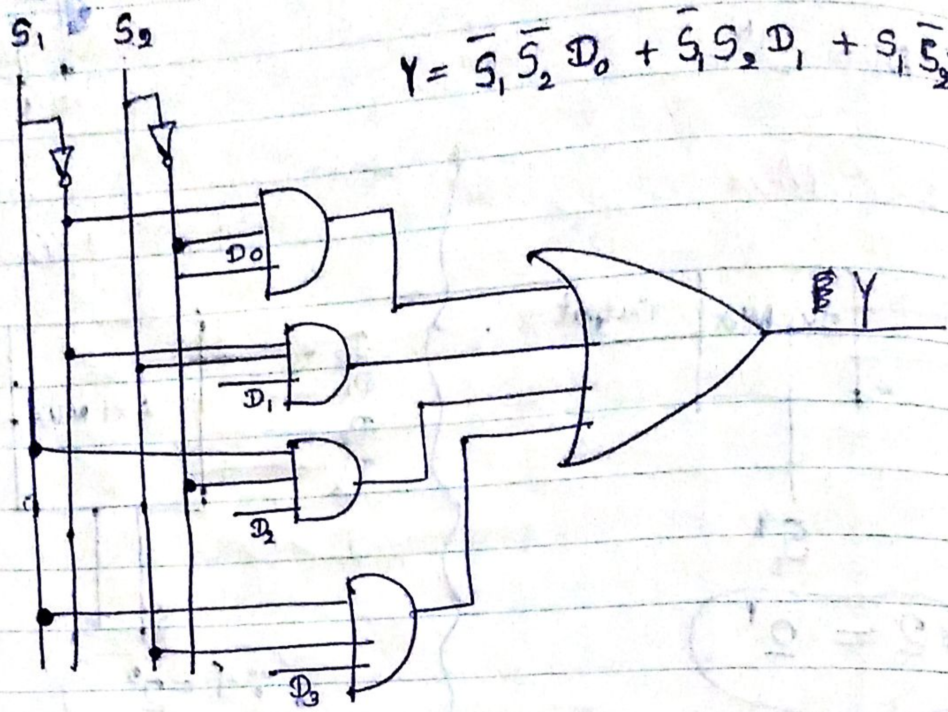


## 4x1 Multiplexer. (Implementation of Bo... function using Mu...

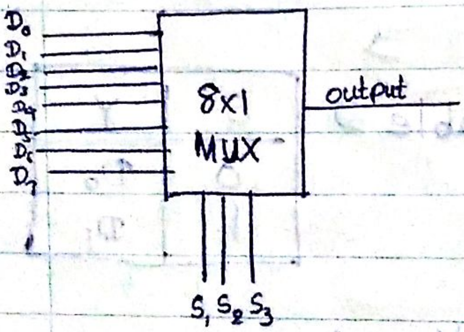



| $S_1$ | $S_2$ | Y | |
|---|---|---|---|
| 0 | 0 | $D_0$ | $\bar S_1 \bar S_2 D_0$ |
| 0 | 1 | $D_1$ | $\bar S_1 S_2 D_1$ |
| 1 | 0 | $D_2$ | $S_1 \bar S_2 D_2$ |
| 1 | 1 | $D_3$ | $S_1 S_2 D_3$ |

$$Y = \bar S_1 \bar S_2 D_0 + \bar S_1 S_2 D_1 + S_1 \bar S_2 D_2 + S_1 \ldots$$



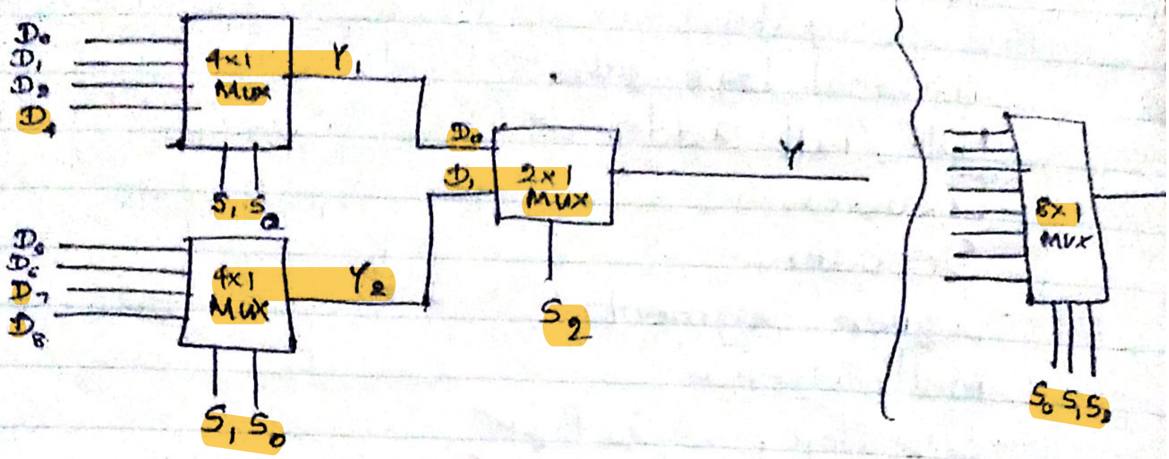

## 8x1 Multiplexer.



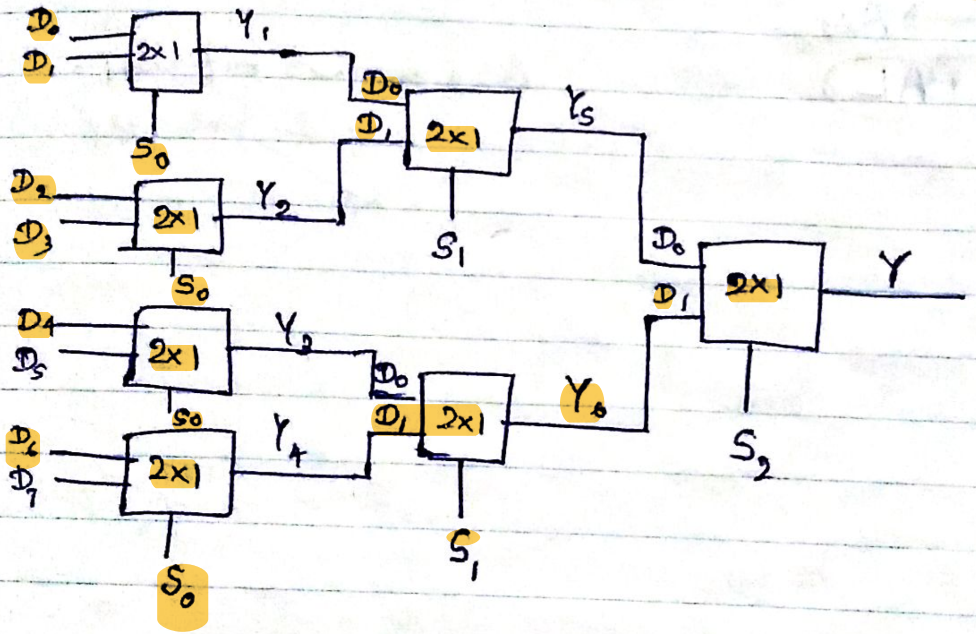

| $S_1$ | $S_2$ | $S_3$ | Y |
|---|---|---|---|
| 0 | 0 | 0 | $D_0$ |
| 0 | 0 | 1 | $D_1$ |
| 0 | 1 | 0 | $D_2$ |
| 0 | 1 | 1 | $D_3$ |
| 1 | 0 | 0 | $D_4$ |
| 1 | 0 | 1 | $D_5$ |
| 1 | 1 | 0 | $D_6$ |
| 1 | 1 | 1 | $D_7$ |

$$Y = \bar S_1 \bar S_2 \bar S_3 D_0 + \bar S_1 \bar S_2 S_3 D_1 + \bar S_1 S_2 \bar S_3 D_2$$
$$+ \bar S_1 S_2 S_3 D_3 + S_1 \bar S_2 \bar S_3 D_4 + S_1 \bar S_2 S_3 D_5$$
$$+ S_1 S_2 \bar S_3 D_6 + S_1 S_2 S_3 D_7$$

## 8×1 Multiplexer using 4×1 Multiplexer




## 8x1 Multiplexer using 2x1 Multiplexer




## 16x1 MUX using 4x1 MUX